

# Soft computing

## Introduction

- Soft computing refers to a group of computational techniques that are based on AI and natural selection.
- It provides cost effective solutions to the complex real life problems.
- Soft computing is a tolerance of following:
  - (1) Imprecise
  - (2) Uncertainty
  - (3) Partial truth
  - (4) Approximation
- Role model of soft computing is human mind.

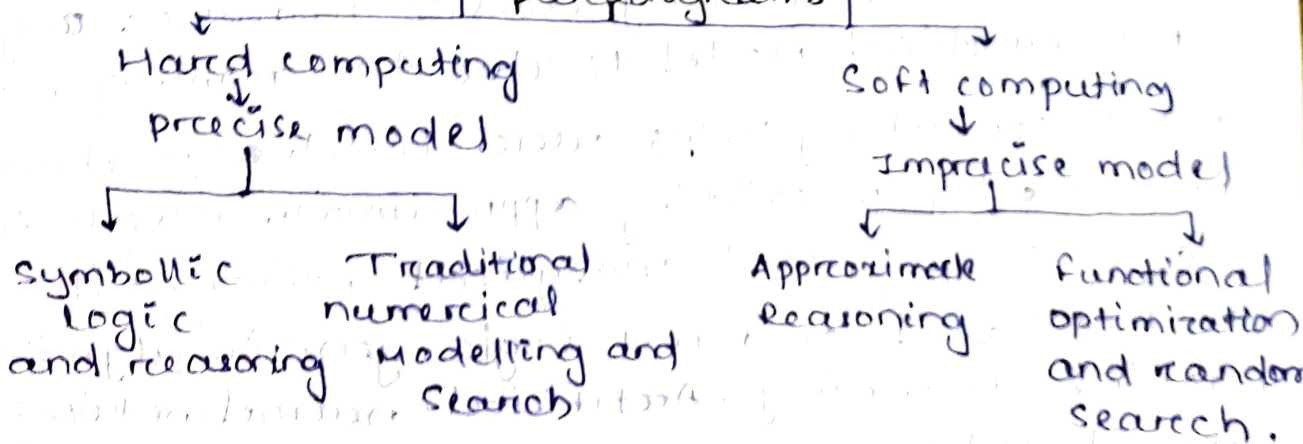
## Hard computing

- (1) Hard computing uses mathematical algorithm approximation in precise to solve certain problems. results.
- (2) It provides a precise and exact solution of the problems.
- (3) More computation time.
- (4) It is mainly based on binary logic and numerical systems.
- (5) Sequential computation.

## Soft computing

- (1) Soft computing allows mathematical algorithm approximation in precise to solve certain problems. results.
- (2) It provides a imprecise solution.
- (3) Less computation time.
- (4) It depends on approximate value.
- (5) Parallel computation.

### Computational paradigms



→ Zadeh coined a term soft computing in 1993.

→ Let us consider a program that actually doesn't have any solution via traditional computing.

→ But soft computing gives the approximate solution.

→  $string1 = "xyz"$

$string2 = "xyw"$

→ As  $string1$  and  $string2$  are same solutions, ~~and~~  $string1$  and  $string2$  are not equal.

→ How much  $string1$  and  $string2$  are same, ~~and~~  $string$  might be 66% similar according to soft computing.

Characteristics of Soft Computing:

→ soft computing provides an approximate but precise solution for real life problems.  
↓  
accurate

→ The algorithm of soft computing are adaptive so the current process is not affected by any kind of change in the environmental.

→ Adaptive (It stores past value also).

→ The concept of soft computing is based on learning from experimental data, it means that soft computing doesn't require any mathematical model to solve the problems.

→ Soft computing helps users to solve real world problems by providing approximate result that conventional and analytical model can't solve.

→ It is based on fuzzy logic, genetic algorithm, machine learning, Artificial neural network, expert system.

Why we need soft computing:-

→ Sometimes conventional computing or analytical models doesn't provide a solution to some real world problems in that case we require soft computing technique to obtain approximate solution.

Application area of soft computing:-

→ soft computing technique is used for handwriting recognition.

→ Image processing and data compression

→ Automotive systems and manufacturing

→ Decision support systems.

→ Speech and vision recognition system

→ washing machine, microwave, Heater and refrigerator → Home appliances.

→ kitchen appliance such as rice cooker

→ Power plant can be control using an intelligent control system.

→ In biomedical applications soft computing technique can be used to solve biomedical problems like diagnosis, monitoring, treatment and therapy.

Elements of soft computing:-

→ The phrase "soft computing" was coined by Lotfi A. Zadeh in the early 1990.

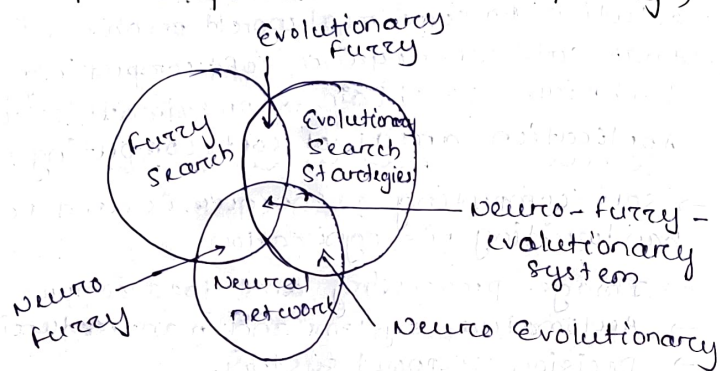
→ The principal components of soft computing includes fuzzy systems (fuzzy set theory, fuzzy logic, fuzzy inference systems etc)

→ Rough set theory (approximate or bounded value), Artificial neural network, probabilistic reasoning, evolutionary learning strategies (including genetic algorithm, simulated annealing, ant colony optimization)

swarm optimization etc.

## classification of soft computing:-

(Principle components of soft computing)



As stated by zadeh, the guiding principle of soft computing is to exploit, the tolerance for imprecision, uncertainty, partial truth and approximation to achieve tractability, robustness, low cost etc.

### Technique

### Application domain

- |                                      |                                                                               |
|--------------------------------------|-------------------------------------------------------------------------------|
| (1) <u>Fuzzy systems</u>             | :- vagueness / imprecision, inexactness, approximate reasoning.               |
| (2) <u>Rough sets</u>                | :- not clearly informed, vagueness / inexactness in information systems.      |
| (3) <u>Artificial neural network</u> | :- Learning and curve fitting pattern classification, association, clustering |
| (4) <u>Probabilistic reasoning</u>   | :- uncertainty and belief propagation                                         |
| (5) <u>Evolutionary search</u>       | :- complex optimization                                                       |

## fuzzy logic:-

- It allow truth value.
- Fuzzy logic is a mathematical framework it is used for
- Fuzzy logic is used to model and control system that has value or imprecise input and output system.
- Fuzzy inference system (FIS) are widely used in various applications including controlling system and decision support system.

## Neural Network:-

- Neural network inspired by the structure of the human brain are use for tasks like pattern recognition, classification, regression and optimization.
- They consists of interconnected nodes/neurons organized in layers.
- Deep learning is the subset of neural network is popular due to its ability to handle large dataset and complex pattern.
- Convolutional neural network and Recurrent neural network (RNN) are used in image recognition and sequential data analysis respectively.

Genetic Algorithms:- It is a optimization technique based on the principle of natural selection and genetics.

- They are used to find approximate solution to optimization and search problems.
- Genetic algorithm maintain a population of potential solutions and use selection, crossover, mutation operators to evolve

better solution to <sup>our</sup> generations.

## → Evolutionary Algorithms:-

\* In addition to genetic algorithms, other evolutionary algorithm includes genetic programming, evolutionary strategies, differential evolution (DE).

\* This technique can be used for optimization, machine learning and evolving algorithms and structures.

## \* Bayesian Networks:-

→ Bayesian networks are probability graphical models use for reasoning under uncertainty.

→ They represent probability relationship between variables using directed acyclic graph.

→ Bayesian networks are apply decision support system, medical diagnosis, risk analysis.

## Swarm Intelligence:-

→ This techniques are based on the birds, ant, bee.

→ Swarm intelligence is adopted from the social organism behaviour such as ant, bee, birds.

→ They include algorithms like birds (particle swarm optimization), Ant colony optimization (ACO), Artificial Bee colony (ABC) algorithms.

→ This algorithms are used for optimization problems including routing, scheduling and clustering.

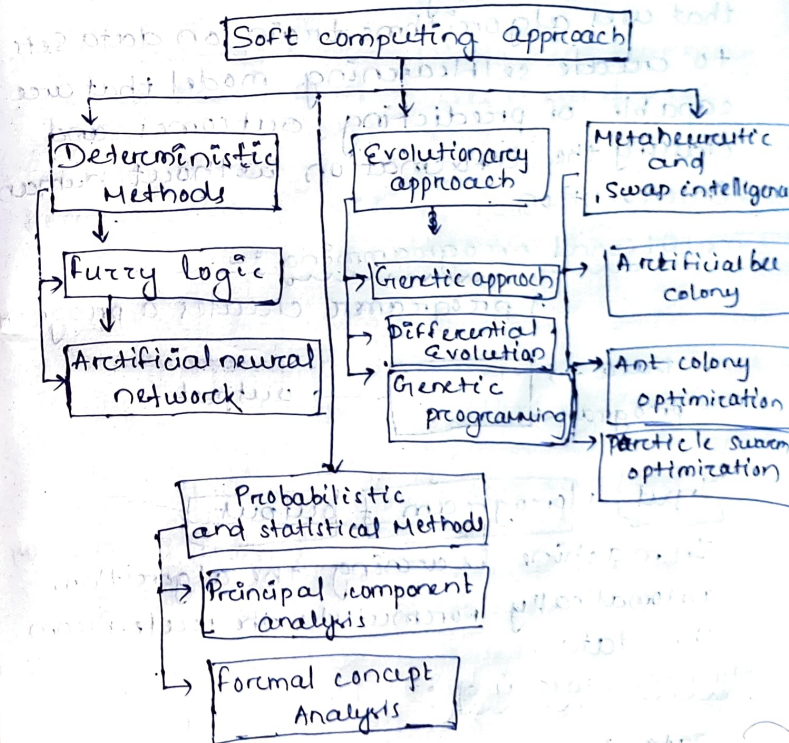
Rough set theory:- It deals with imprecise and incomplete information, it provides a formal framework to handle uncertainty and make decision based on discernibility and indiscernibility.

→ It is using data preprocessing and feature selection.

## Artificial Immune System:-

→ Bayesian network, Artificial immune system inspired by the human immune system.

→ Artificial immune systems are used for anomaly detection and pattern recognition.



## Machine learning:-

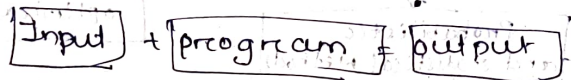
→ Machine learning is defined as a technology that is used to train machines to perform various actions such as prediction, estimation etc based on historical data and past experience.



→ Machine learning is a subfield of AI that uses algorithms to train on data sets to create self-learning models that are capable of predicting outcomes and classify the information without human intervention.

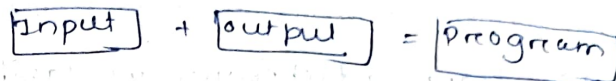
### Traditional programming:-

A programmer creates a program



In machine learning the algorithm automatically formulates the rule from the data.

### Machine learning:-



### Difference between Machine learning vs Traditional programming:-

#### Traditional programming

(1) In traditional programming rule-based code is written by the developer depending on the problem statements.

(2) Traditional programming is rule based and deterministic. It has no self learning features like ML & AI.

(3) Traditional programming is totally depends on the intelligence of the developer so it has very limited capability.

(4) Traditional programming is used to build application and software systems that have specific functionality.

#### Machine learning

(1) In machine learning it is subset of AI, that focus on learning from data to develop an algorithm that can be used to make a prediction.

(2) Machine learning uses data driven approach, it is typically train on historical data and then use to make prediction on new data.

(3) Machine learning can find patterns and insights in large datasets, that might be difficult for humans to discover.

(4) Machine learning is the subsets of AI and now it is used in various AI based task like chatbot, question-answering, self-driven car etc.

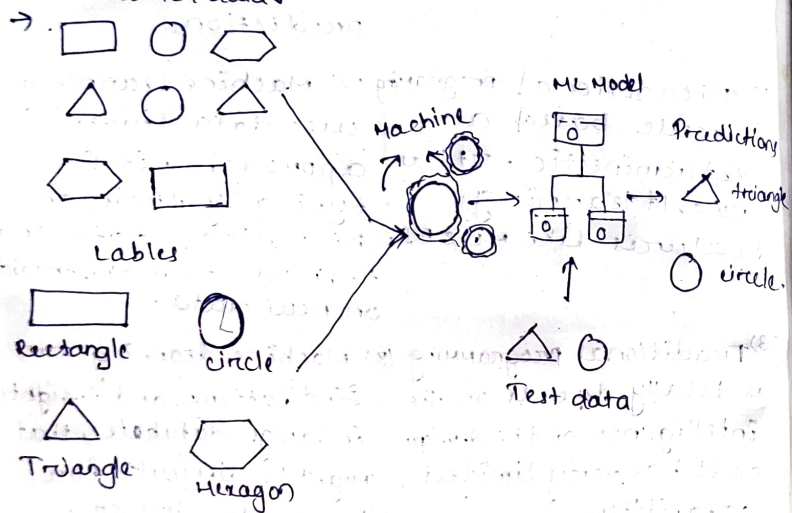
## Classification of Machine Learning Techniques:

Machine learning techniques are classified into 4 types:-

- (1) Supervised learning
- (2) Unsupervised learning
- (3) Semi Supervised learning
- (4) Reinforcement.

### Supervised Learning:-

→ In supervised learning sample labeled data are provided to the ML system for training and then system then predict the output based on the training data.



→ Images and shapes are the input to the ML Model and labels of those images as shape name are the output data.

→ Based on these input and output data the model <sup>learned</sup> ~~learned~~ to predict the category of unseen image data wheather a rectangle, circle, triangle or hexagon.

→ Supervised learning can be group further into 2 categories of algorithms:-

- (1) classification. (discrete)
- (2) Regression. (contiguous)

→ classification is a type of supervised learning where the algorithm learns to assign input data to a specific category or class, based on input features.

→ The output labels in classification are discrete values.

→ classification algorithms are binary or multiple.

→ Regression is a type of supervised learning where the algorithm learns to predict continuous value based on input features.

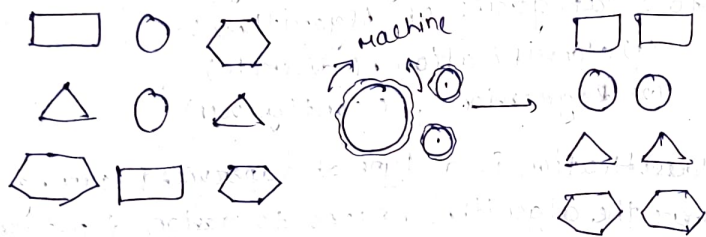
→ The output labels in regressions are contiguous value such as stock price, housing price etc.

### Unsupervised Machine Learning:-

→ Unsupervised learning is a learning method in which a machine learns without any supervision.

→ The training is provided to the Machine with the set of data that hasn't been labeled, classified or categorized, and the algorithm needs to act on that data without any supervision.

→ The goal of unsupervised learning is to restructure the input data into new features or a group of objects which with similar patterns.



### (Unsupervised Learning)

→ In unsupervised learning we don't have predetermined results.

→ The basic tries to find useful insights from the huge amount of data.

→ unsupervised learning classified into 2 categories of algorithm:-

(1) clustering

(2) Association.

#### Clustering:-

→ clustering algorithms groups similar data-points together and based on their character.

→ The goal is to identify groups or cluster data points that are similar to each other and distinct from ~~every~~ other groups.

#### Association:-

→ Association rule on supervised learning method which is used for finding relationship between variables in the large database.

→ It determines set of items that occur together in a database.

e.g:- People who buy a item (suppose a computer) and also tend to purchase y items (i.e software)

### Semi-Supervised Learning:-

→ Semi-supervised learning lies in between supervised and unsupervised learning.

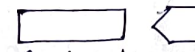
→ It is a Mixture of labeled and unlabeled data with larger proportion of unlabeled data.

→ It uses the combination of labeled and unlabeled datasets during that training period.

→ Input data



Partial labels



Rectangle Hexagon

Machine learning Models



Prediction

It is a triangle

unlabeled data



### (Semi supervised learning)

→ Here the machine first classify the new triangle shape image is not a rectangle or hexagon

→ speech analysis, image classification, Text document classifier

### Reinforcement Learning:-

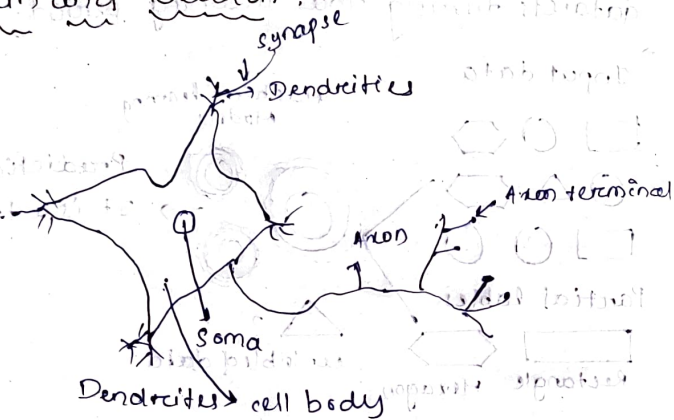
Reinforcement learning is a feedback based learning method in which learning agent gets a rewards for each right action and gets a penalty for each wrong action.

→ The agent <sup>learns</sup> automatically with its feedback and improve its performance.

→ In reinforcement learning the agent interact with the environment and explores it.

- The goal of an agent is to get most reward point and hence it improves the performance.
- e.g.:- traffic control, Marketing and advertising.

### → Structure and functioning of Biological Brain and Neuron:-



### Basic components of Biological Neurons:-

- The majority of neurons encode their activations or outputs as a series of brief electrical pulses.
- The neuron's cell body (soma) processes the incoming activations and convert them into output activation.
- The neurons nucleus the genetic material in the form DNA.
- Dendrites are emanates from the cell body and provide the receptive zones that receives activation from other zones.

→ The human brain consists of huge no. of neurons approximately  $10^{11}$  with numerous interconnection.

→ Each neuron is connected about  $10^4$  other neurons.

→ The biological neurons consists of 3 other main parts:- (i) Dendrites:- Receives signal from neighbouring neurons.

(ii) Soma or cell body:- Accumulates the signals receives through the dendrites.

(iii) Axon:- Transmits signals from soma to the action neurons.

→ Axon terminals propagate to neighbouring neurons.

→ Functions:-

→ The dendrites collect stimuli from the neighbouring neurons and passed it on to soma which is the main body of the cell.

→ The soma accumulates the stimuli receives through the dendrites.

→ It fires when sufficient stimuli is obtained.

→ When a neuron fires it transmits its own stimuli from the Axon.

→ The stimulus passes on to the neighbouring neurons through the Axon terminals.

→ There is a small gap between the end of an axon terminals and the adjacent dendrites of the neighbouring neurons, this gap is called synapse.

→ A nervous neuron is a electric impulse.

→ it is transmitted across a synaptic gap by means of electro-chemical process.

- The synaptic gap has a important role to play in the activity of the nervous system.
- It scales the input signal by a weight. if the input signal is  $x$ , and the synaptic weight is  $w$ , then the stimulus that finally reaches the soma due to input  $x$  is the product  $x \times w$ .

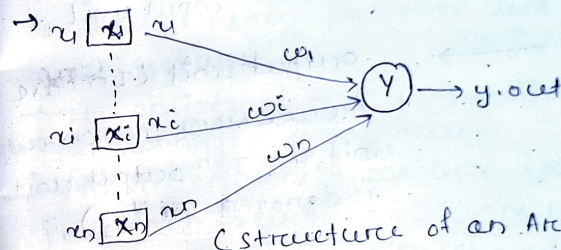
→ The significance of the weight " $w$ " provided by the synaptic gap lies in the fact that together with other synaptic weights, including embody the knowledge store in the network of neurons.

### → The ~~sim~~ features of the biological neuron

- \* The body of the neuron is called soma that act as a processing element to receive many signals to the dendrites simultaneously.
- \* The strengths of the incoming signals are modified by the synaptic gaps.
- \* The role of the soma i.e. processing element of a neuron is simple.
- \* It sums up the weighted input signals. If the <sup>sum</sup> is sufficiently high it transmits an output signal to the axon.
- \* The output signal reaches the receiving neuron in the neighbourhood to the axon terminals.
- \* The weight factor provided by the synaptic gaps are modified over time and experience.

### Model of an Artificial neuron:

- An Artificial neuron is a computational model based on the structure and functionality of a biological neuron.
- It consists of a processing element, a number of inputs and weighted edges connecting each inputs to the processing element.



- A processing unit is usually represented by a circle as integrated by the unit  $Y$ .
- An Artificial neuron may consists of a number of inputs units.

→  $x_1, x_2, \dots, x_n$  are the input units.

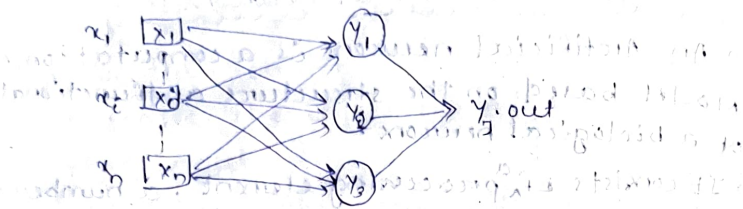
→ The corresponding input signals are  $w_1, w_2, \dots, w_n$  and  $Y.out$  is the output signals of the processing unit  $Y$ .

→ The net input  $Y.in$  to the processing element  $Y$  is obtained as  $Y.in = x_1 w_1 + x_2 w_2 + \dots + x_n w_n$

$$= \sum_{i=1}^n x_i w_i$$

→ If there are more than one output unit then their net input unit to the  $j$ th output unit  $Y_j$  denoted as  $Y.in_j$  is,

$$Y.in_j = x_1 w_{1j} + x_2 w_{2j} + \dots + x_n w_{nj} = \sum_{i=1}^n x_i w_{ij}$$



| <u>Symbol used</u> | <u>Description</u>                                                                                                                                                                                                                                  |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $x_i$              | → The $i$ th input unit.                                                                                                                                                                                                                            |
| $y$                | → output unit (In case there are more than one output unit, the $j$ th output unit is denoted as $y_j$ .)                                                                                                                                           |
| $x_i$              | → Signal to the input unit $x_i$ . (This signal is transmitted to the output unit $y$ scaled by the weight $w_i$ .)                                                                                                                                 |
| $w_i$              | → The weight associated with the interconnection between input unit $x_i$ and the output unit $y$ .<br>(In case there are more than one outputs unit $y_j$ denotes the weight bet <sup>n</sup> input unit $x_i$ and the $j$ th output unit $y_j$ .) |
| $y_{in}$           | → The total input to the output unit $y$ . (It is a algebraic sum of all weighted input to the $y$ .)                                                                                                                                               |

## Symbol used

$y.out$



## Description

Signal transmitted by the output unit  $y$ . (It is known as the activation of  $y$ .)

## Characteristics of brain:-

- The most important feature of brain is its extremely parallel and this centralized architecture.
- It consists of more or less 100 billion neurons interconnected among them.
- The interconnections are local in nature that each neurons are connected to its neighbors there is no centralized control in brain.
- The neurons act on the basis of local information.
- Each neurons function in parallel mode and concurrently.

## Features of Brain:-

Architecture:- The average human brain consists of about 100 billion neurons.

- There are nearly  $10^{15}$  numbers of interconnections among these neurons, hence the brains architecture is highly connected.

Mode of operation:- The brain operates the extreme parallel mode.

speed:- processing is slow as compare to speed of the digital system.

fault tolerance:- The brain is highly fault tolerant. knowledge is store within the brain in distributed manner.

Storage Mechanism:- The brain stores information as strength of the interconnections among its neurons.

→ New information can be added by adjusting the weights, without destroying the already stored information.

→ Control:- There is no global control in the brain.

→ A neuron act on local information available with its neighbouring neurons.

Comparison between ANN and Biological neural network

Artificial neuron n/w      Biological neuron Network

Speed:- ANNs are faster → BNS are slow in processing information.

Processing:- Many programs → BNS can perform massively parallel operation and they operate in a sequential mode one instruction after another on a conventional computer. → The brain process the capacity to operate the massive parallel each of them having only few steps.

Size and complexity:- → neural networks has large no. of computing elements (10<sup>5</sup>). The number of neurons in the brain is estimated about 10<sup>10</sup> or 10<sup>15</sup>. The brain performs complex pattern recognition task which can not be recognize on a computer.

Storage:- In a computer → neuron n/w store the information are stored information in this in memory which is addressed strength of the by its location, any new interconnection. Information in the same in the brain is adaptable.

ANN location destroy the old information. → because new information is added by adjusting the interconnection without destroy the old information.

fault tolerance:-

ANN are inherently not fault tolerant since the information concepted in the memory can't be retrieved. → They exhibit fault tolerance since the information is distributed in the connection throughout the n/w.

→ control Mechanism  
There is a control unit which monitors all the activities of the computing.

→ There is no central control for processing information in the brain.

Concept of learning / training:

- The main property of the artificial neuron network is its capability to learn.
- Learning / training is a process by means of which a neuron network adapt itself to a stimulus by making proper parameter adjustments, resulting in the production of desired response.
- There are 2 types of learning in ANN.
  - parameter learning
  - structure learning

Parameter learning

It update the connecting weights in a neural network.

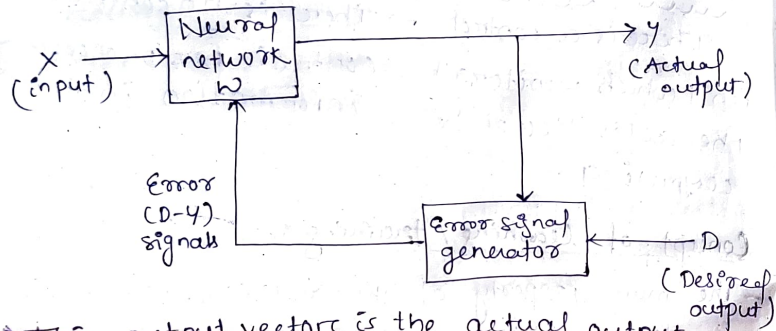
## Structure learning

It focus on the changing network structure which include the no. of processing element & their connection type.

- ANN can broadly classify into 3 categories
- Supervised learning
  - ~~semi~~ Unsupervised learning
  - Reinforcement learning

## Supervised learning

The learning is performed with the help of feature during training the input vector is presented to the network which result in a output network.

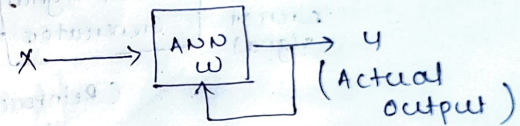


- This output vector is the actual output vector.
- Then the actual output vector is compare with the desired or target output vector.
- If there exist a difference between the 2 output vector (Y & D) then the error signal is generated by the network.
- This error signal is used for adjustment of weights until the actual output matches the desired output.

→ In this type of learning/training a supervised feature is required for error minimization.

→ In supervised learning, it is assumed that the correct target output values are known for each output pattern.

## Unsupervised learning:-

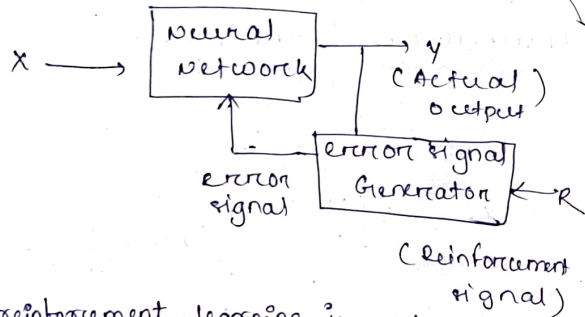


(unsupervised learning)

- It is performed without the help of a teacher.
- In this training process, the network receives the input pattern and organizes this pattern to perform cluster.
- when a new input pattern is applies, the neural network is output response indicating that class to which the input pattern belongs.
- If for an input, a pattern class can not be found when a new class is generated.
- In unsupervised learning, there is no feedback from the environment to inform what the output should be or wheathere the output are correct or not. In this case diagram, The network it self discover a pattern or categories from the input data and relation from the input data over the output.

→ This process is called self organizing in which exact clusters will be formed by discovering similarities and dissimilarities among the objects.

### Reinforcement learning:-



→ The reinforcement learning is a form of supervised learning because the network receives some feedback from its environment.

→ However, the feedback obtained is only evaluative & non-instructive.

→ The external reinforcement signals are processed in the critic signal generator & the obtained critic signals are sent to the ANN for adjustment of weights properly. So, as to get better feedback in future.

→ The reinforcement learning is also called learning with critic.

### Transfer / Activation functions:-

→ The activation function is applied over the net input to calculate the output of ANN.

→ The function that maps the net input values to the output signal value i.e. the activation is known as the activation

function of the unit.

### Types of activation function:-

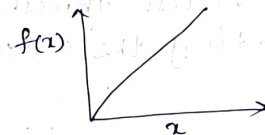
#### Identity function:

→ It is a linear function & the identity function passes on the incoming signals as the outgoing signal without any change.

→ The identity activation function can be defined as

$$f(x) = x \quad \forall x$$

here, the output remains same as input.



Identity func

### Binary threshold signal function:-

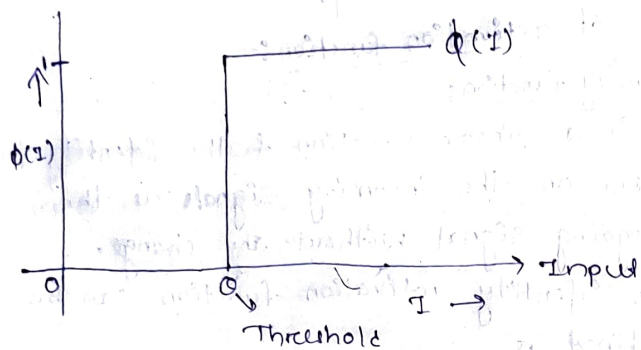
→ Neurons with binary signal function are referred to as binary neurons or threshold logic neurons.

→ In this case the sum is compared with threshold value (0).

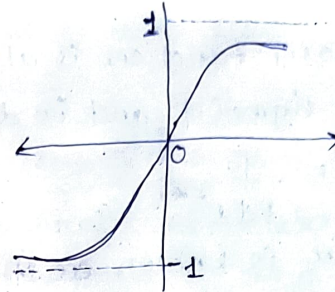
→ If the value of  $I$  (net input) is greater than (0), then the output is equal to one otherwise the output is 0.

$$\phi(I) = \begin{cases} 1 & \text{if } I > 0 \\ 0 & \text{otherwise} \end{cases}$$

→ The thresholding function is also known as step function or heavy-side function



(Thresholding fun<sup>n</sup>)



(Bipolar sigmoid function)

$$g(x) = \frac{1}{1 + e^{-\alpha x}}$$

$$g'(x) = 0 - \frac{d}{dx} (1 + e^{-\alpha x})$$

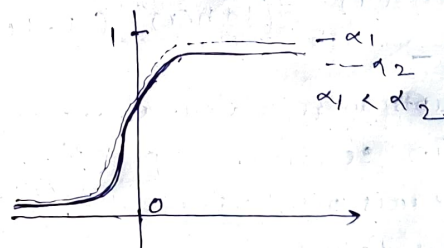
$$= \frac{-(-\alpha e^{-\alpha x})}{(1 + e^{-\alpha x})^2}$$

$$= \frac{\alpha e^{-\alpha x}}{(1 + e^{-\alpha x})^2}$$

$$= \frac{\alpha e^{-\alpha x}}{(1 + e^{-\alpha x})^2}$$

→ Depending on the requirement <sup>binary sigmoid function can be scaled to any range of values given for an application.</sup>

→ The mostly widely used range is -1 to +1 any corresponding sigmoid function is referred to as bipolar sigmoid function.



(Binary sigmoid function)

→ Mostly single layer net use thresholding function for calculating the output from the net input.

The sigmoid function:-

→ The step function is not contiguous, it is not differentiable.

→ Some ANN training algorithms required the activation function be contiguous and differentiable.

→ The step function is not suitable for some cases.

→ The sigmoid function is a contiguous function that varies between the asymptotic values 0 and 1 or -1 to 1, and is given by  $\phi(x) = \frac{1}{1 + e^{-\alpha x}}$

$$1 + e^{-\alpha x}$$

where  $\alpha$  is the slope parameter.

→ Binary sigmoid function is also referred to as logistic sigmoid, and is defined by the  $g(x) = \frac{1}{1 + e^{-x}}$

→ The parameter is known as stepness parameter.

→ The derivative  $g'(x)$  is expressed as -

$$g'(x) = \frac{d}{dx} g(x) = g(x)(1 - g(x))$$

→ Hyperbolic tangent function:-

\* The hyperbolic tangent function is closely related to bipolar sigmoid function

\* When the net input data is binary and not continuously valued in the range of 0 to 1.

\* ~~When the net input is binary and not continuous valued in the range 0 to 1~~ then convert to bipolar form and the bipolar sigmoid or hyperbolic tangent function is applied by the processing unit.

$$h(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$h'(x) = (1 - h(x)) (1 + h(x))$$

Perceptron:-

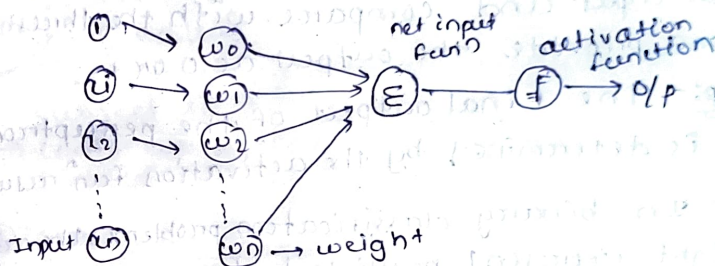
A perceptron is an neural network unit that contains certain computation to detect features or business intelligence in the input data.

→ It is a function that maps its input "x" which is multiply by the learned weight

coefficient and generates an output value  $f(x)$ .

→ It is Basic components of perceptron:-

Mr. Frank Rosenblatt invented the perceptron model as a binary classifier



Input feature:-

→ The perceptron takes multiple input features each input feature represents a characteristic or attribute of the input data

weight:- Each input feature associated with a weight determining the significance of each input feature in influencing the perceptron.

→ During the training the weight are adjusted to learn the optimal value.

Summation function:- The perceptron calculates the weighted sum of its input using summation function.

→ The summation function combines the inputs with a respective weights to produce a weighted sum.

## Activation function:-

- The weighted sum is then passed through an activation function.
- The perceptron uses heavy side step function, which takes a sum value as input and compare with the threshold and provide the output as 0 or 1.

O/p:- The final output of the perceptron is determined by the activation fun<sup>n</sup> result.

e.g:- In binary classification problem, the output might represent predicted class 0 or 1.

Bias:- The bias allows the model to make adjustments that are independent to the I/p. → It is additional parameter i.e. learn during training.

- In neural network, bias can be divided to the constants which is adopted to the product of feature weight.
- It helps the model to shift the activation function towards the +ve or -ve side.

## Types of perceptron:-

There are 2 types of perceptron.

Such as:- (1) single layer

(2) Multi layer.

### single layer perceptron:-

It can learn only linearly separable pattern.

### multilayer perceptron:-

- It can learn about 2 or more layers having a greater processing power.
- This model also known as back propagation algorithm which enters 2 stages such as

forward stage, backward stages.

## Binary & continuous input:-

### Binary input:-

- It takes only 2 values which are 0 & 1.
- It can be used as a representation of categorical variables. e.g:- Male, female.

### Continuous input:-

- It refers to input data that has an infinite no. of values.
- These inputs are usually measurable of subject to change over time.
- Continuous input values can be subdivided into parts.
- e.g:- high, weight, temperature.

### Discrete inputs:-

- It refers to numeric data that are finite and countable.
- These inputs usually only take on certain values.
- e.g:- Number of students in the class.
- Discrete inputs can also be used to represent categorical data.
- Discrete inputs are usually non-negative or +ve numbers and they are countable.

## Linear Separability:-

A linearly separable set of patterns is one that can be completely partition by a decision plane into two classes.

- In perceptron for a given set of linearly separable patterns it is always possible to find a perceptron that solves the corresponding classification problem.
- e.g:- Two sets of points are:

$$A = \{a, b, c\} = \{(-1, 3), (1, 2), (3, 3)\}$$

$$B = \{p, q, r\} = \{(0, -1), (2, 1), (3, 0)\}$$

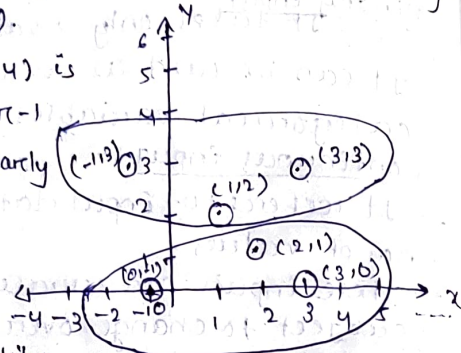
→ If B is contain (3, 4).

→ If the point (3, 4) is

present in cluster-1

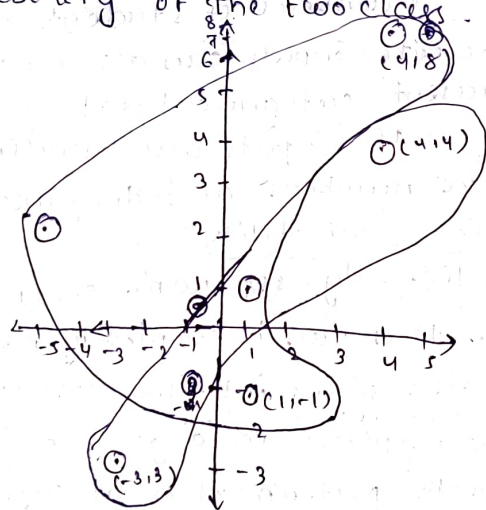
then they are linearly

non separable.



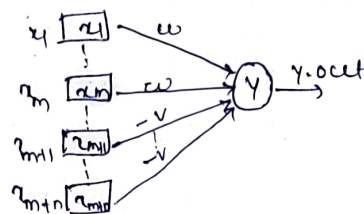
→ Linear separability problems can be separated using linear decision boundary.

e.g consider the dataset with two features  $x_1$  and  $x_2$  in which the points  $(-1, 1), (1, 1), (-3, -3), (4, 4)$  belongs to class  $x_1$  and the points  $(-1, 1), (1, -1), (-5, 2), (4, 8)$  belongs to class  $x_2$ . Find the linear separability of the two class.



Linearly non-separable

## The MCCLOCH - PITTS Neural Model:-



(Structure of a MCCulloch-PITTS Neuron)

→ The architecture consists of number of input units that is  $x_1, x_m, x_{m+1}, \dots, x_{mn}$  connected to a single output unit.

→ The interconnecting links are unidirectional

→ There are 2 kinds of input units :-

1) Excitatory

2) Inhibitory.

→ Excitatory inputs are  $x_1, x_2, \dots, x_m$ ,  
Inhibitory units are  $x_{m+1}, x_{mn}$ .

→ The excitatory inputs connected to the output unit through positively weighted links.

→ The inhibitory inputs have -ve weight on their connecting path to the output unit.

→ All excitatory weights have the same positive magnitude  $w$  and all inhibitory weights have the same <sup>negative</sup> weight magnitude  $-v$ .

→ The activation  $y_{out} = f(y_{in})$  is binary.  
i.e either 0 or 1.

→ The activation function is a binary step function. It is 1 if the net input  $y_{in}$  is greater than or equal to a given threshold value 0.

and 0 otherwise.

## Unit - 2